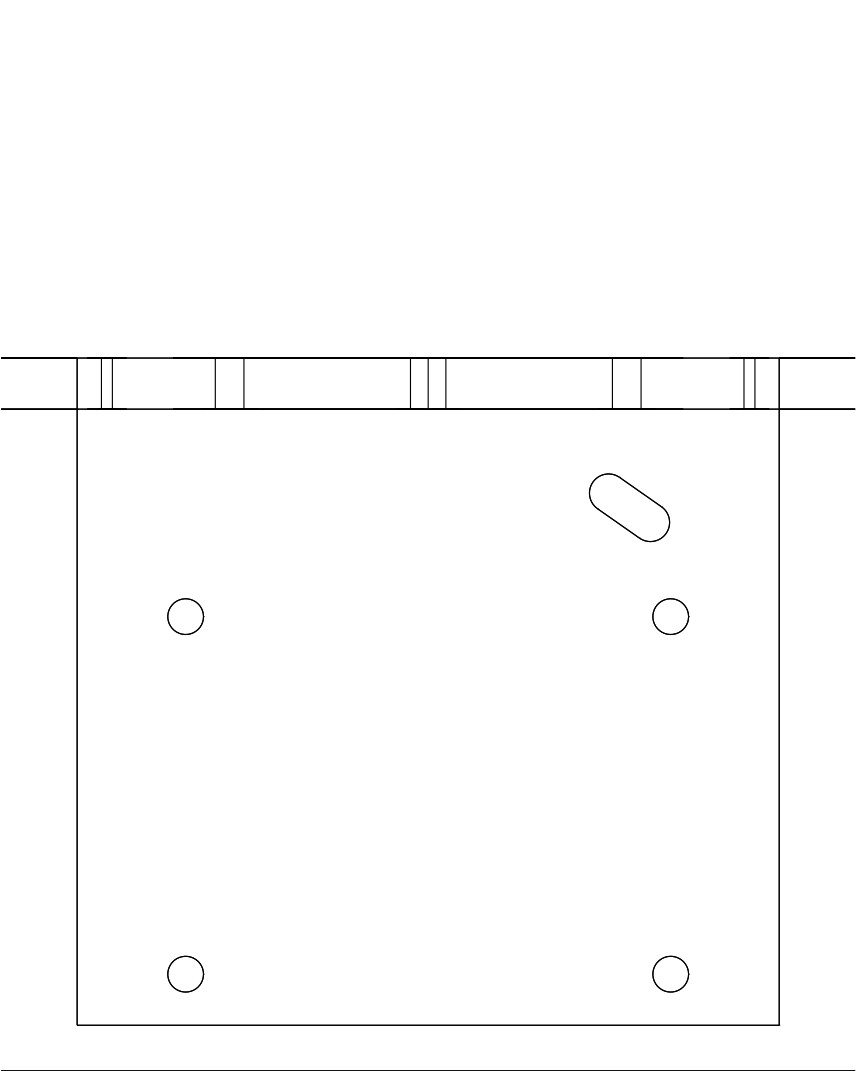


A16-S101-root-to-shoulder-L

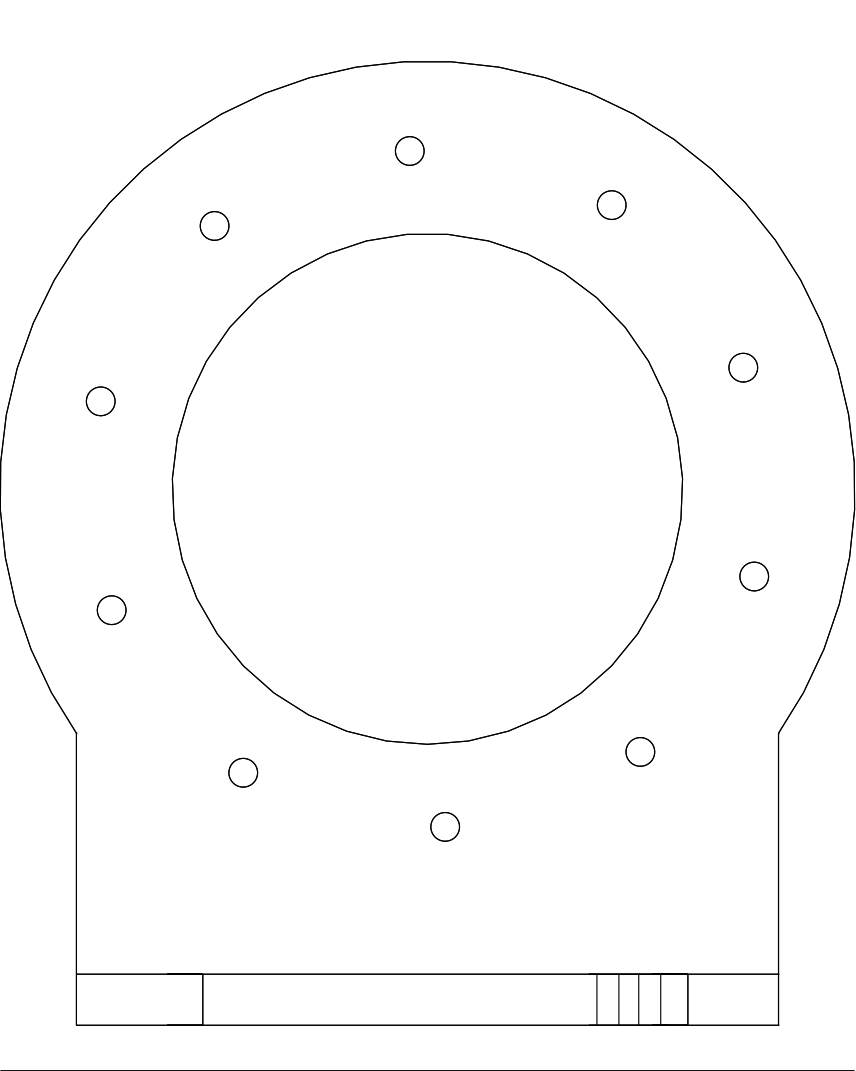
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



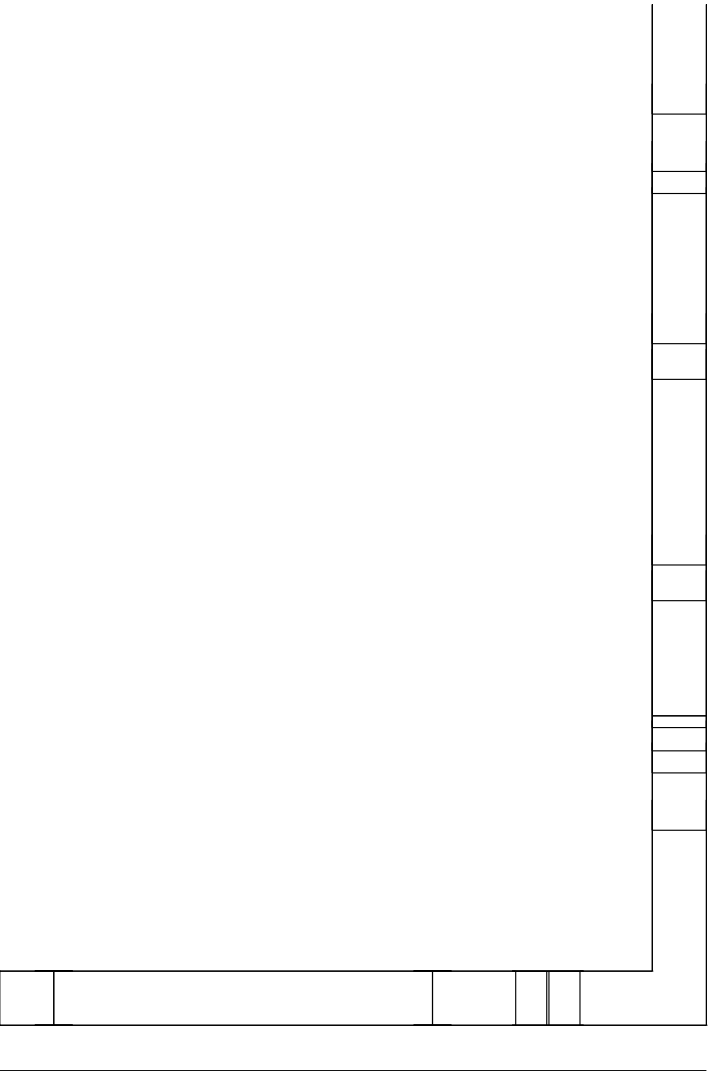
133.838 overall / drawing scale0.844
Vertical104.500; CAD axes XY

XZ all edges, hidden included



133.838 overall / drawing scale0.844
Vertical150.895; CAD axes XZ

YZ all edges, hidden included



104.500 overall / drawing scale0.895
Vertical150.895; CAD axes YZ

Foot8 thick: X-55..55; Y-36..68.5; Z30..38.
4xD5.6: X+/-38,Y+/-28, parallelZ.
J2 fixed annulus OD134 ID80, faceY68.5, Z centre114.
8-thick annulus, native10xD4.5 RS04 front holes.
Wire capsule6x14,endR3,centreR55 at55deg;235deg port lies outside foot.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

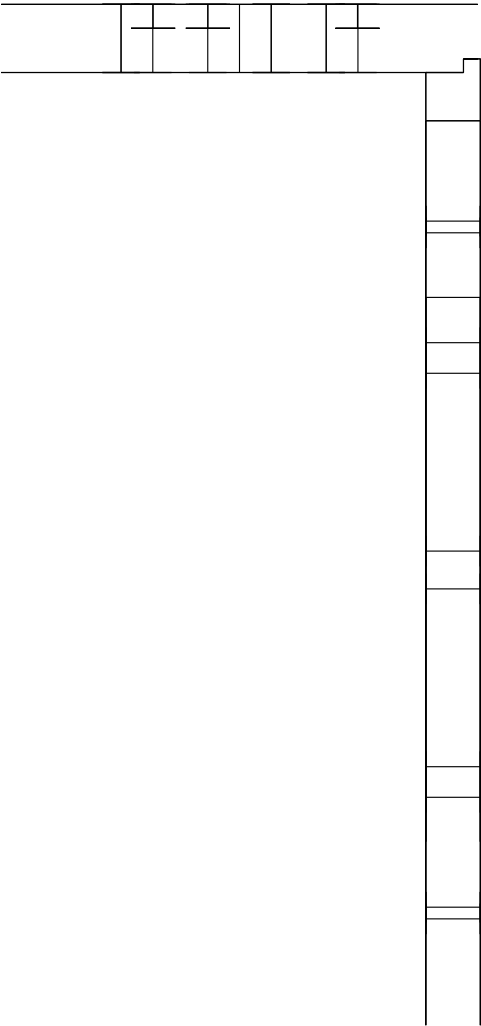
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S102-shoulder-cross-adapter

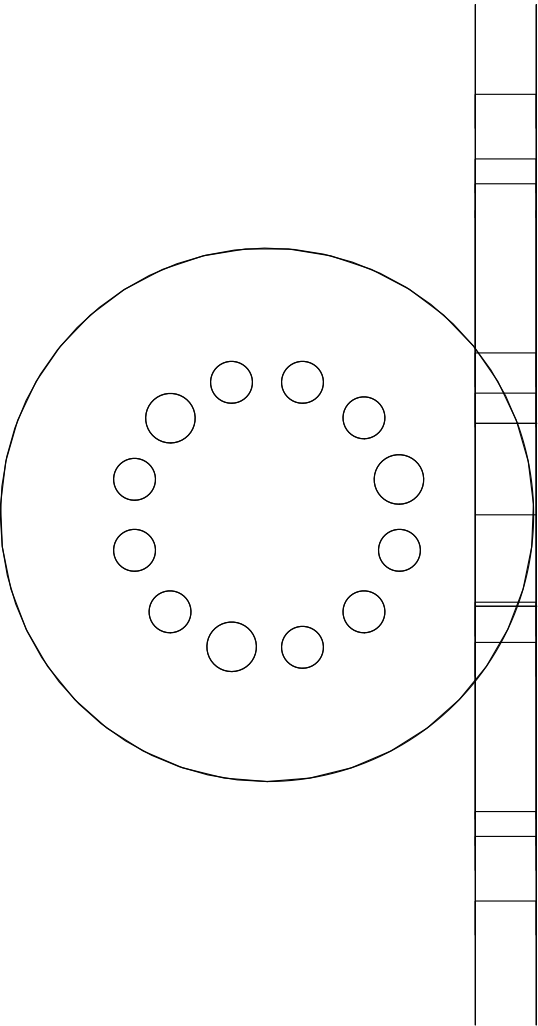
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



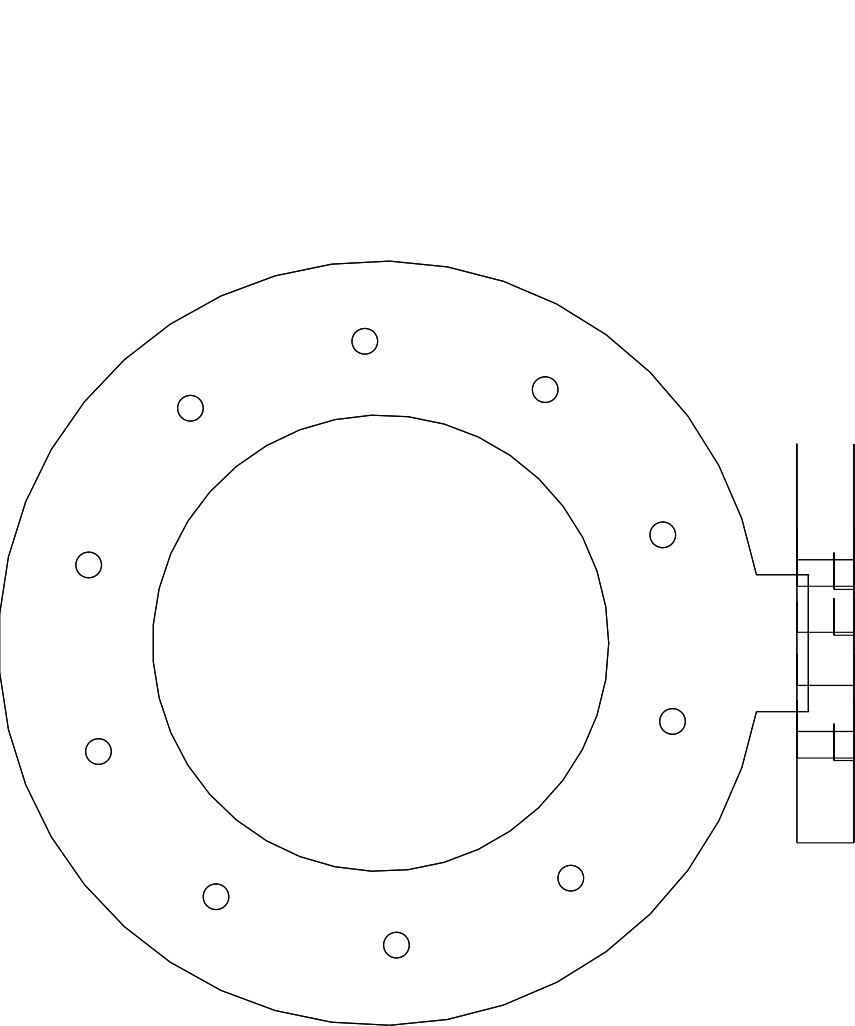
70.350 overall / drawing scale0.901
Vertical149.807; CAD axes XY

XZ all edges, hidden included



70.350 overall / drawing scale1.008
Vertical133.962; CAD axes XZ

YZ all edges, hidden included



149.807 overall / drawing scale0.754
Vertical133.962; CAD axes YZ

J2 output faceY68;10-thick,OD70; native9xD5.5.
J3 annulus shiftedY-15: OD134 ID80,8thick.
Native J3 front10xD4.5;3xD6.5x3.6 locating-pin reliefs.
Output /fixed tool boresD10 /D9.5; native point table attached.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

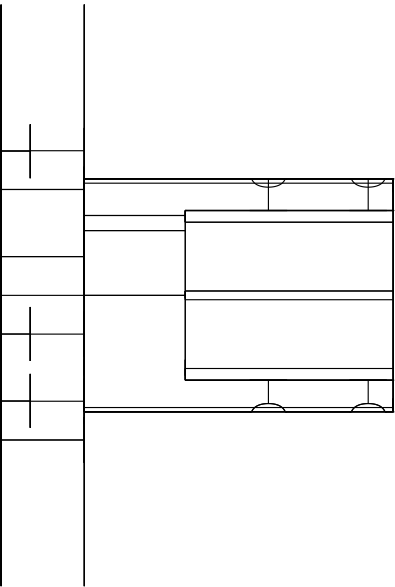
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S103-upper-proximal-socket

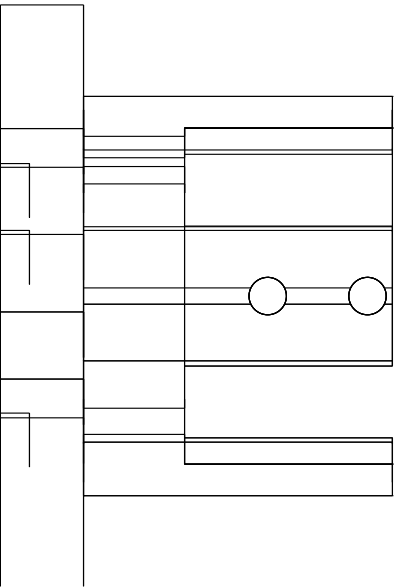
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



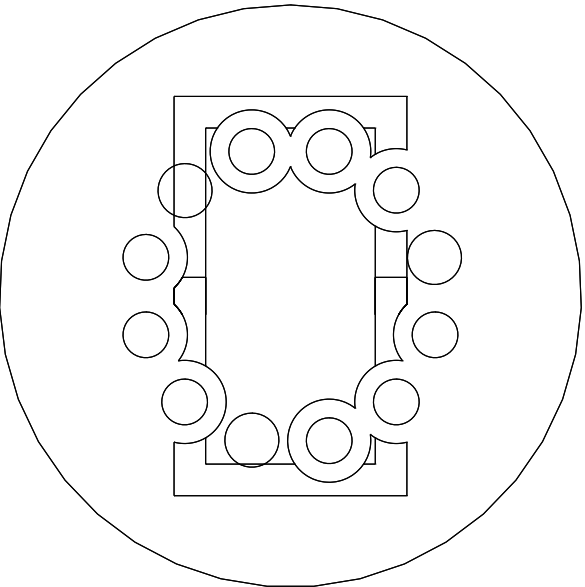
47.150 overall / drawing scale1.100
Vertical69.943; CAD axes XY

XZ all edges, hidden included



47.150 overall / drawing scale1.100
Vertical69.887; CAD axes XZ

YZ all edges, hidden included



69.943 overall / drawing scale1.100
Vertical69.887; CAD axes YZ

Native J3 output faceX27.85; OD70 x10.
SocketX50..75.1; nominal20.4x40.4 internal.
Stock tube startsX50; clamp centresX60/72, Z0.
2xD4.5 parallelY; M4x35 with OD8/ID4.5/L16 crush sleeves.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

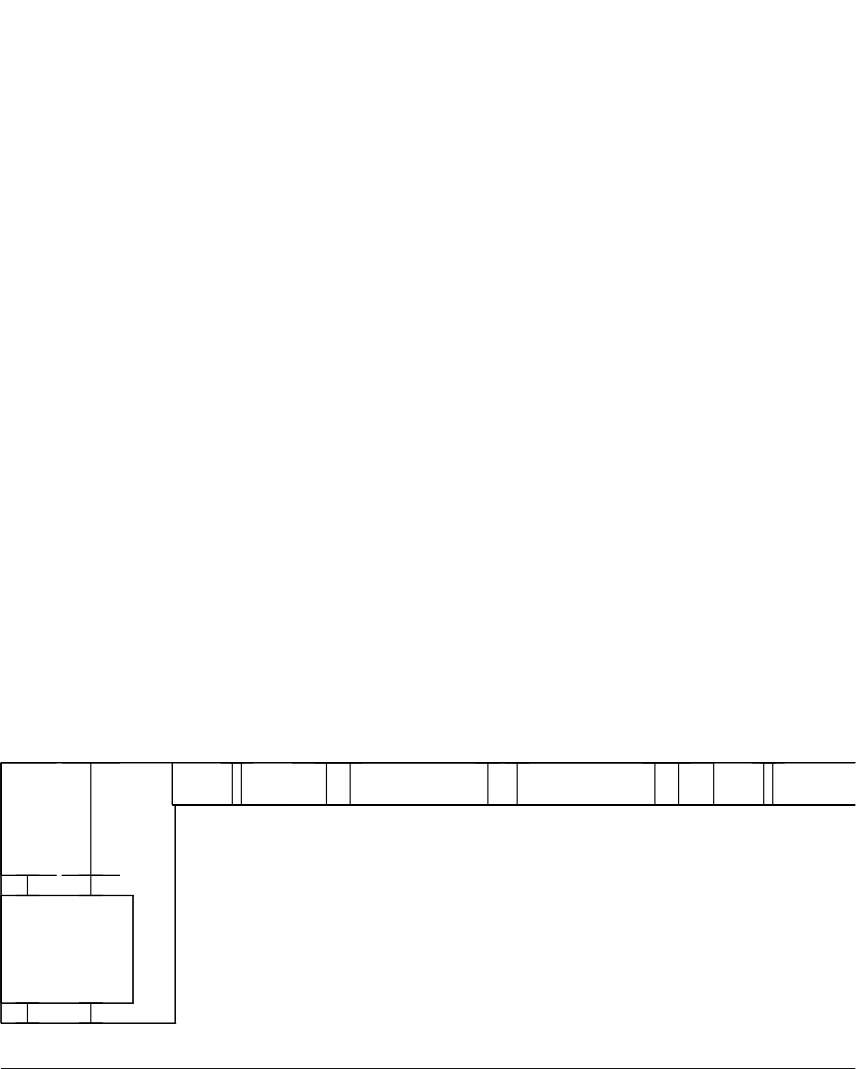
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S104-upper-distal-socket

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



161.830 overall / drawing scale0.698
Vertical49.350; CAD axes XY

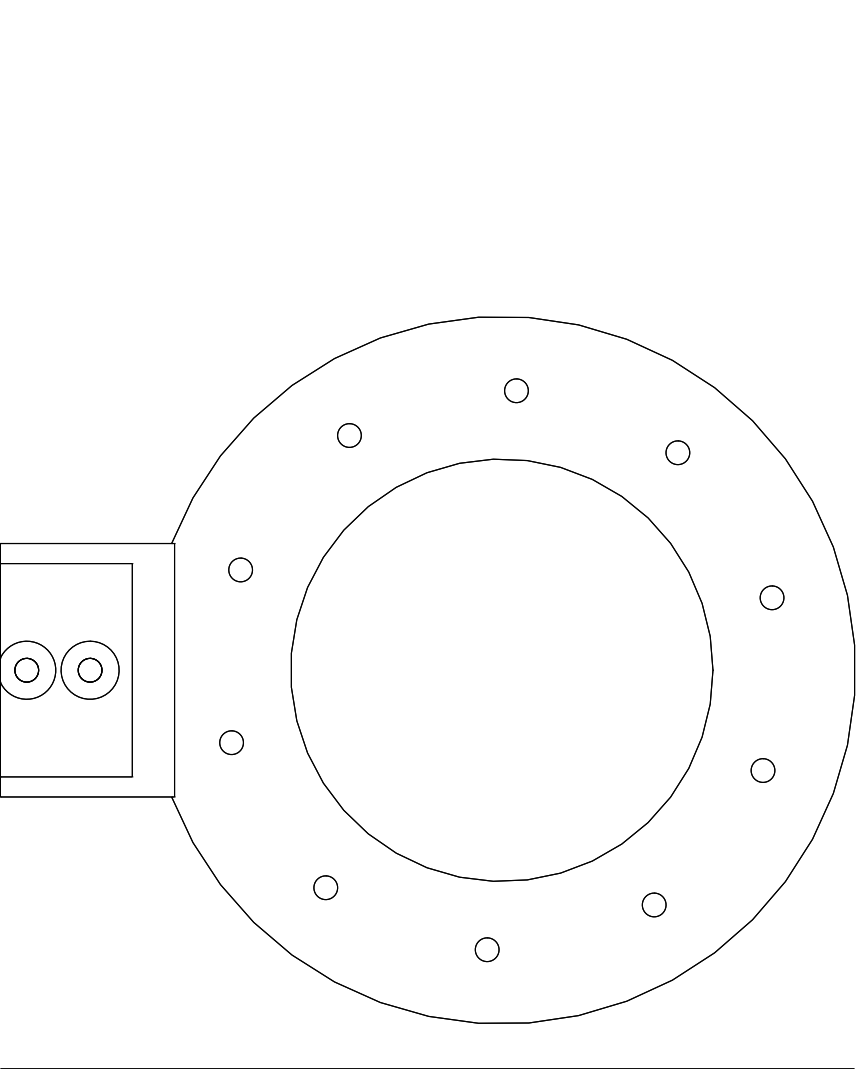
Native J4 fixed faceY27.35, centreX340;8 thick.
SocketX244.9..270;20.4x40.4; tube endsX270.
ClampX250/262,Z0; D4.5 parallelY.
D11 open tool reliefs fromY14 towards+Y,36deep.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

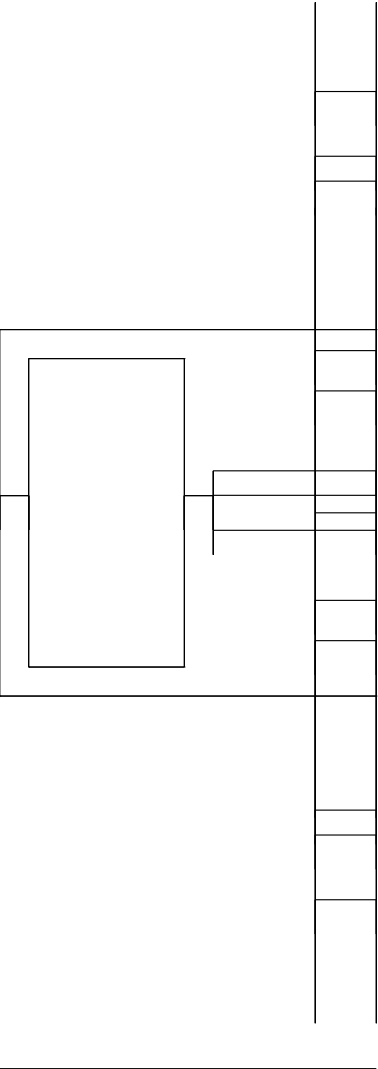
Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

XZ all edges, hidden included



161.830 overall / drawing scale0.698
Vertical133.709; CAD axes XZ

YZ all edges, hidden included



49.350 overall / drawing scale1.010
Vertical133.709; CAD axes YZ

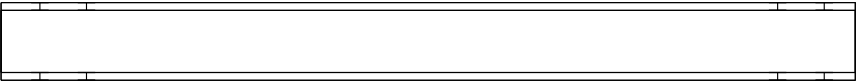
A16-S105-upper-stock-tube

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

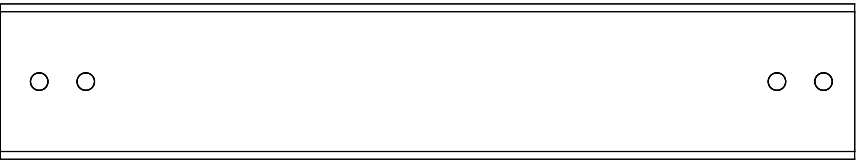
XY all edges, hidden included

XZ all edges, hidden included

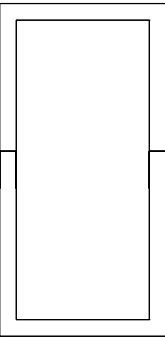
YZ all edges, hidden included



220.000 overall / drawing scale0.514
Vertical20.000; CAD axes XY



220.000 overall / drawing scale0.514
Vertical40.000; CAD axes XZ



20.000 overall / drawing scale1.100
Vertical40.000; CAD axes YZ

Purchased rectangular6061-T6 tube20x40x2, L220.
4xD4.5 through the20mm width; holesX10/22/200/212 from cut end.
Hole axisparallelY; Z0 (20 from bottom40mm face).
8x nominal material surfaces: verify stock dimensions and squareness.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

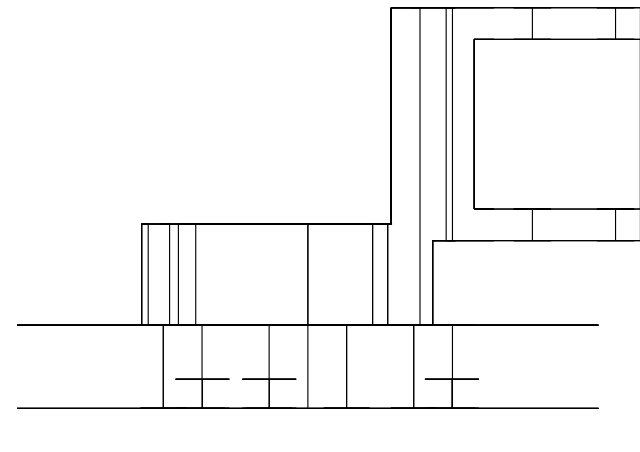
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S106-fore-proximal-socket

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



74.972 overall / drawing scale1.100
Vertical48.150; CAD axes XY

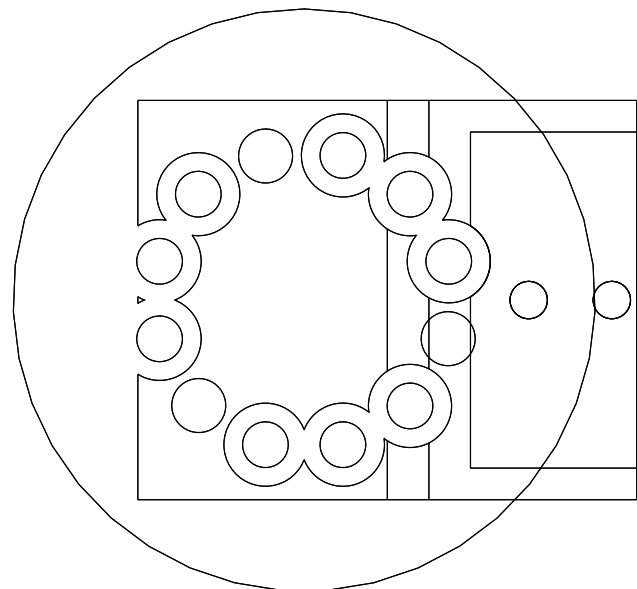
Native J4 output faceY27.85; OD70x10.
Tube socketX20..40.1,20.4x40.4, centreY62.
ClampsX27/37,Z0,D4.5 parallelY.
Native9xD5.5 and3xD6.5x3.6 pin reliefs.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

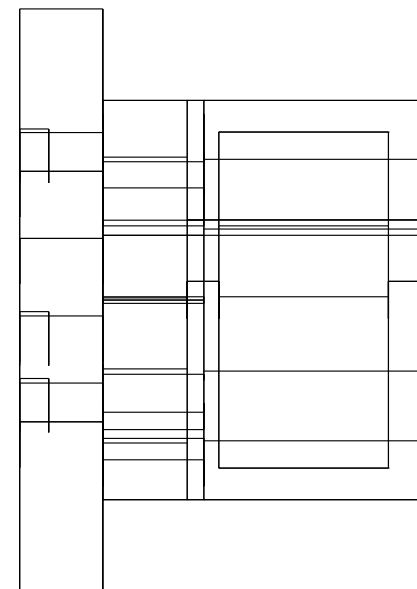
Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

XZ all edges, hidden included



74.972 overall / drawing scale1.100
Vertical69.887; CAD axes XZ

YZ all edges, hidden included



48.150 overall / drawing scale1.100
Vertical69.887; CAD axes YZ

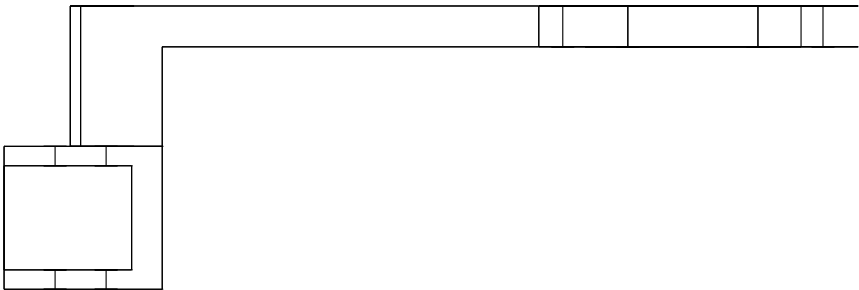
A16-S107-fore-distal-socket

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

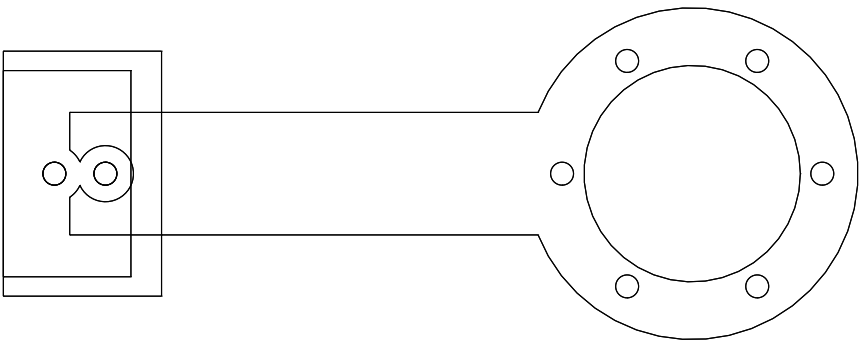
XY all edges, hidden included

XZ all edges, hidden included

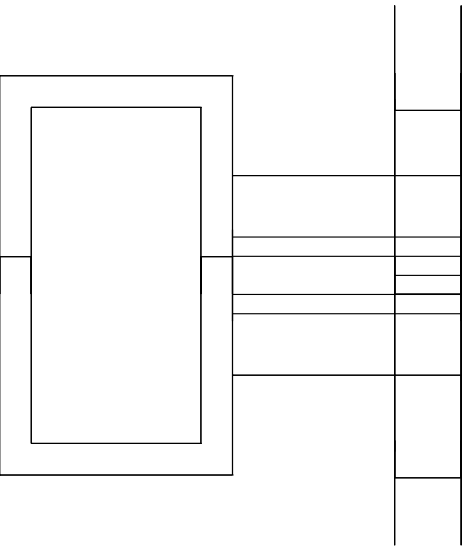
YZ all edges, hidden included



167.418 overall / drawing scale0.675
Vertical55.500; CAD axes XY



167.418 overall / drawing scale0.675
Vertical64.887; CAD axes XZ



55.500 overall / drawing scale1.100
Vertical64.887; CAD axes YZ

Rear-spine conventionalU bracket; J5 fixed centreX185,Y62.

Tube socketX49.9..75; nominal20.4x40.4, centreY62.

ClampsX60/70,Z0,D4.5 parallelY; D11 tool reliefsY76..116.

J5 front ringOD65 ID42.4;8thick; spineY95.5..103.5.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

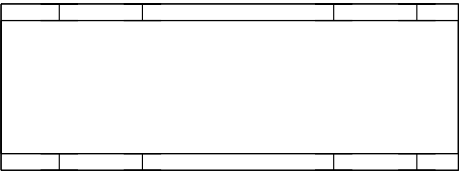
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S108-fore-stock-tube

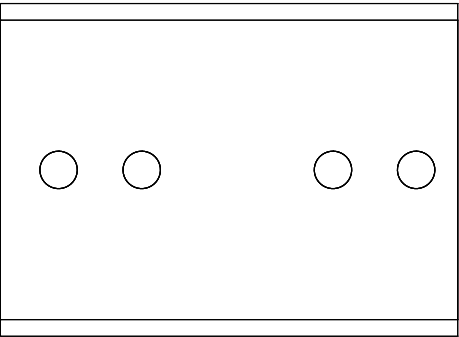
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



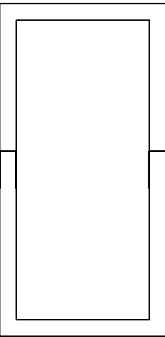
55.000 overall / drawing scale1.100
Vertical20.000; CAD axes XY

XZ all edges, hidden included



55.000 overall / drawing scale1.100
Vertical40.000; CAD axes XZ

YZ all edges, hidden included



20.000 overall / drawing scale1.100
Vertical40.000; CAD axes YZ

Purchased rectangular6061-T6 tube20x40x2, L55.
4xD4.5 through20mm width; X7/17/40/50 from cut end.
Hole axisparallelY; Z0 (20 from bottom40mm face).
Hole/end spacings close: check actual drill burr and edge quality.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

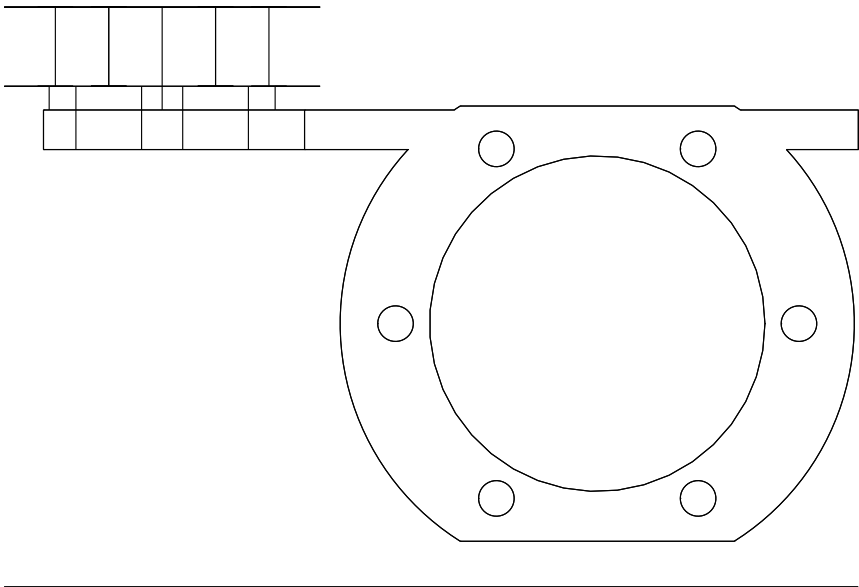
A16-S109-wrist-pitch-yaw-L

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

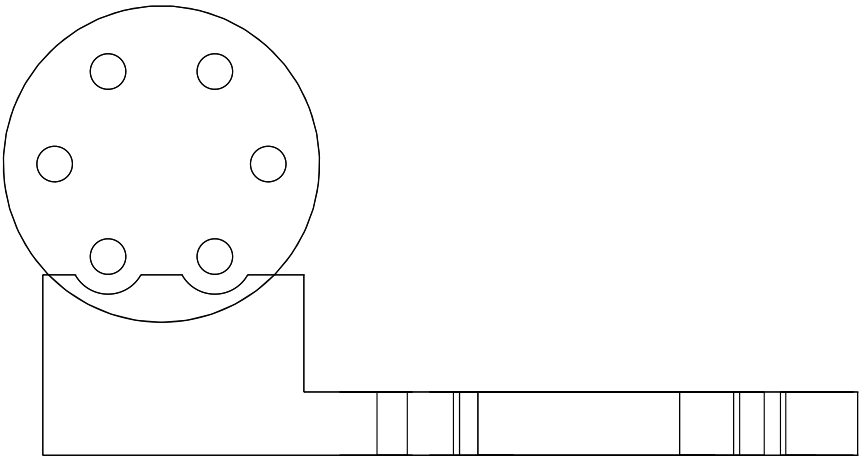
XY all edges, hidden included

XZ all edges, hidden included

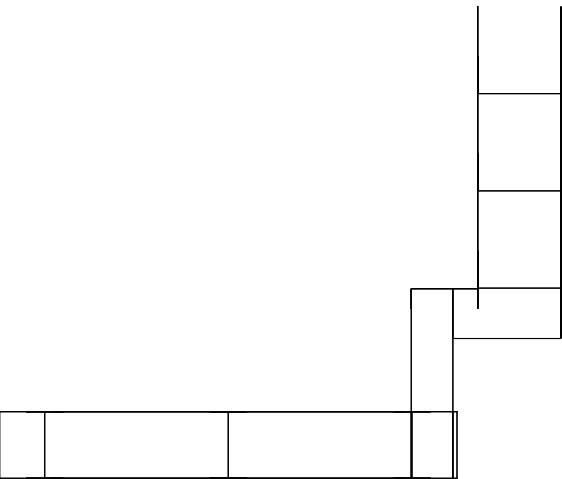
YZ all edges, hidden included



107.984 overall / drawing scale1.046
Vertical67.500; CAD axes XY



107.984 overall / drawing scale1.046
Vertical56.764; CAD axes XZ



67.500 overall / drawing scale1.100
Vertical56.764; CAD axes YZ

Native J5 output faceY40;OD40x10.
J6 fixed planeZ-28.8; centreX55, ringOD65 ID42.4.
Ring clippedY+/-27.5; flat connecting spineY22..27.
6xD4.5 each native RS10P pattern; D9.5 tool reliefs.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

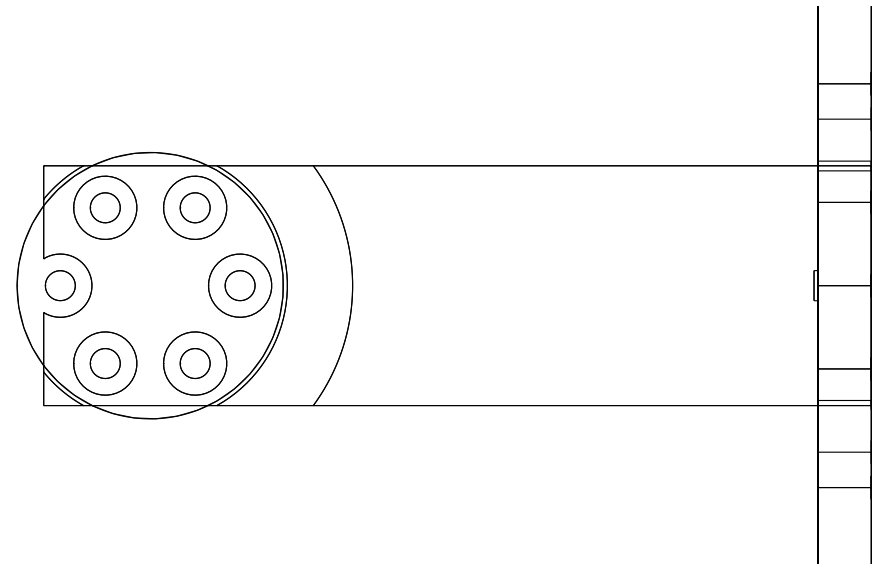
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A16-S110-wrist-yaw-roll-L

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



128.300 overall / drawing scale0.881
Vertical83.956; CAD axes XY

Native J6 output planeZ-30.3;OD40,10 nominal screw grip.
J7 fixed faceX100.3,8thick; centreX75.
J7 annulusOD84 ID37.4;6nativeD3.5 /PCD50.
Cage6xD3.5 PCD70,30+60*k; D8x0.6 washer-edge relief.
OutputD9.5 counterboresZ-40.3..-48.4; strapZ-48.3..-38.3.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

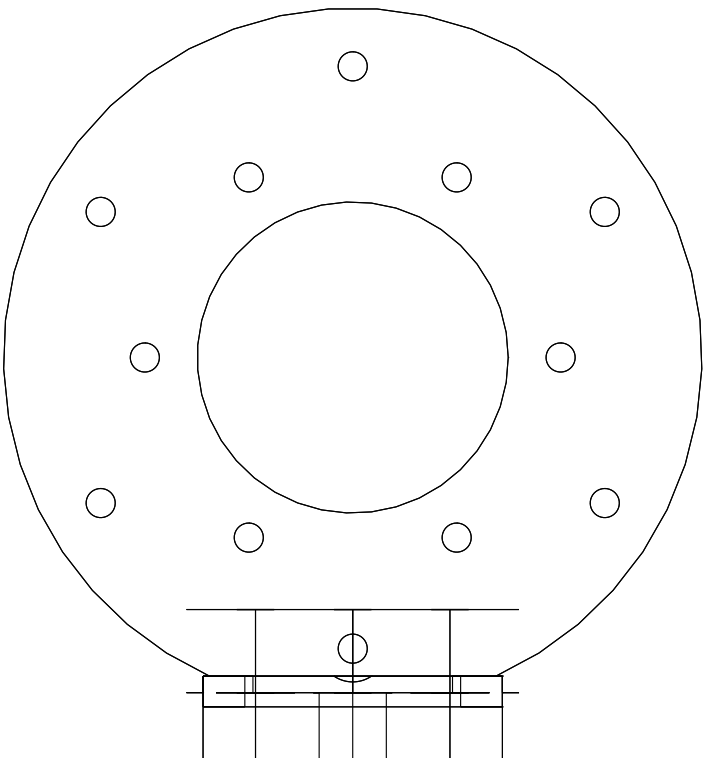
Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

XZ all edges, hidden included



128.300 overall / drawing scale0.881
Vertical90.198; CAD axes XZ

YZ all edges, hidden included

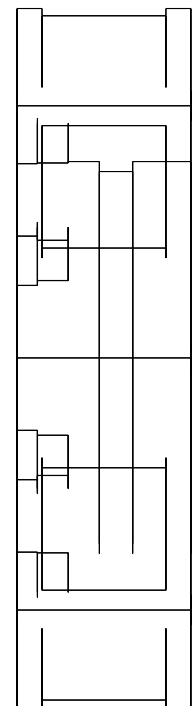


83.956 overall / drawing scale1.100
Vertical90.198; CAD axes YZ

A13-J7-101-bearing-housing

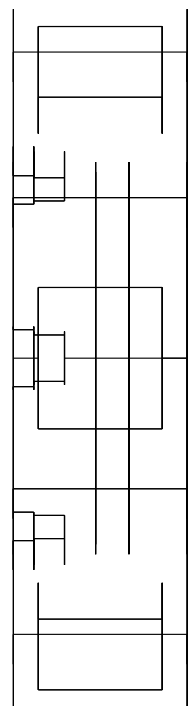
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



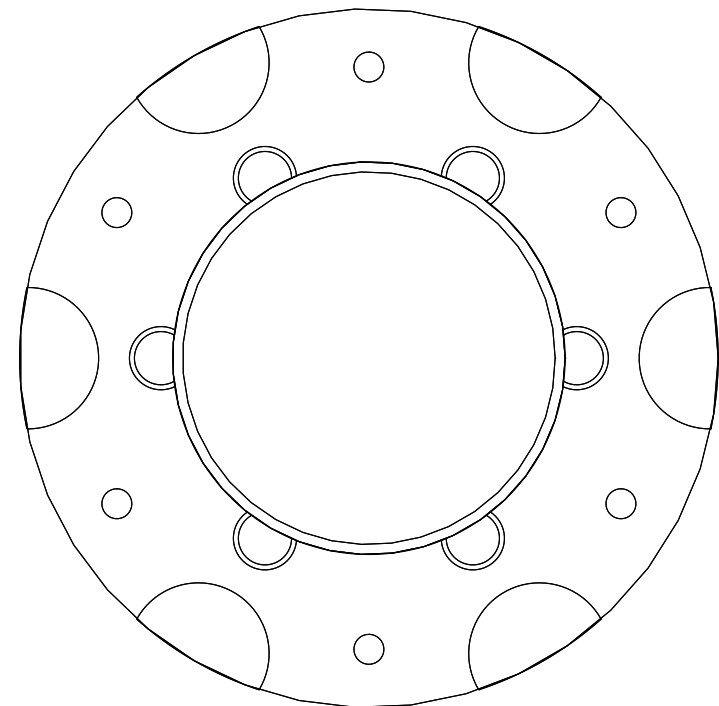
20.900 overall / drawing scale1.100
Vertical83.999; CAD axes XY

XZ all edges, hidden included



20.900 overall / drawing scale1.100
Vertical83.932; CAD axes XZ

YZ all edges, hidden included



83.999 overall / drawing scale1.100
Vertical83.932; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A13-J7-102-bearing-retainer

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



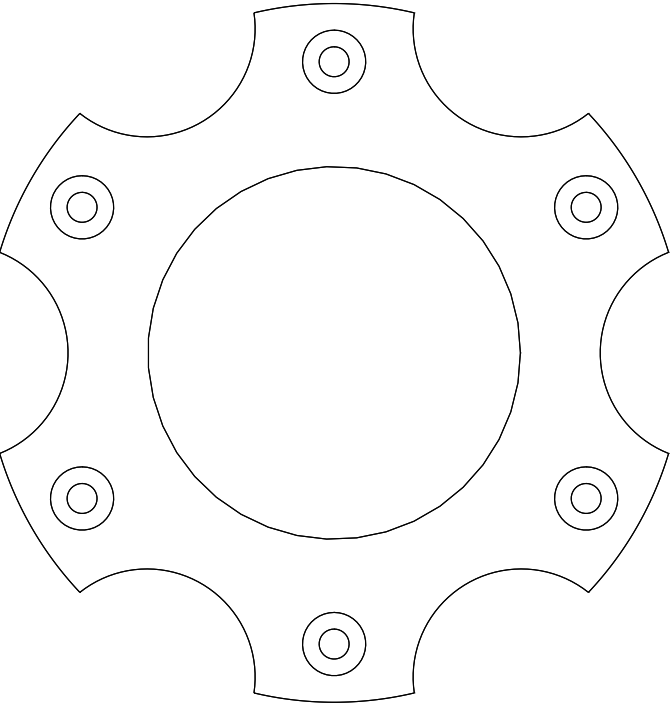
2.500 overall / drawing scale1.100
Vertical80.444; CAD axes XY

XZ all edges, hidden included



2.500 overall / drawing scale1.100
Vertical83.999; CAD axes XZ

YZ all edges, hidden included



80.444 overall / drawing scale1.100
Vertical83.999; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

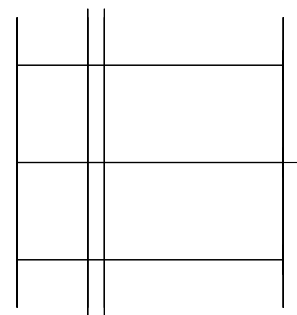
A13-J7-103-output-journal

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

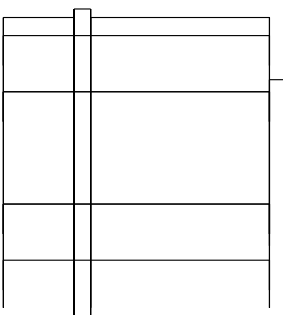
XY all edges, hidden included

XZ all edges, hidden included

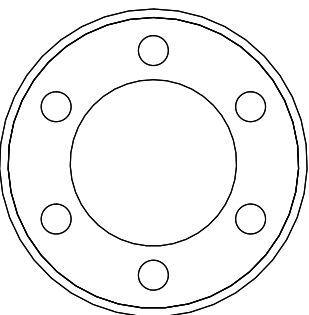
YZ all edges, hidden included



34.000 overall / drawing scale1.100
Vertical36.970; CAD axes XY



34.000 overall / drawing scale1.100
Vertical36.940; CAD axes XZ



36.970 overall / drawing scale1.100
Vertical36.940; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A13-J7-104-inner-spacer

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included

XZ all edges, hidden included

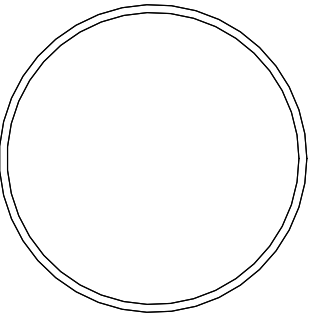
YZ all edges, hidden included



3.500 overall / drawing scale1.100
Vertical36.940; CAD axes XY



3.500 overall / drawing scale1.100
Vertical36.970; CAD axes XZ



36.940 overall / drawing scale1.100
Vertical36.970; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

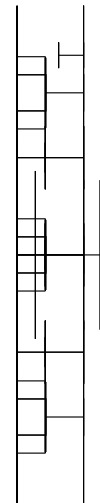
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A13-IF-107-piloted-flange

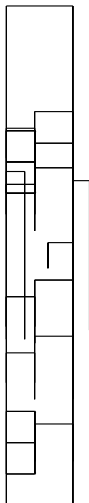
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



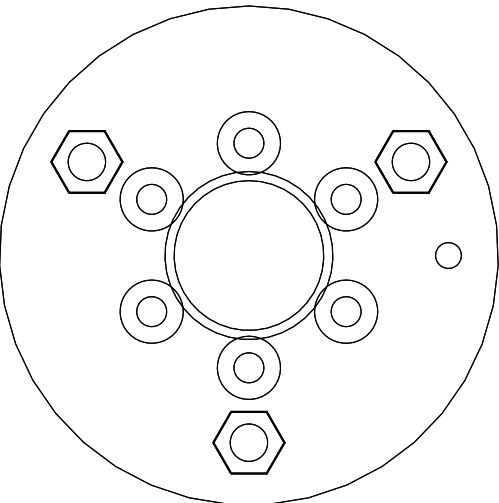
10.000 overall / drawing scale1.100
Vertical59.951; CAD axes XY

XZ all edges, hidden included



10.000 overall / drawing scale1.100
Vertical59.903; CAD axes XZ

YZ all edges, hidden included



59.951 overall / drawing scale1.100
Vertical59.903; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

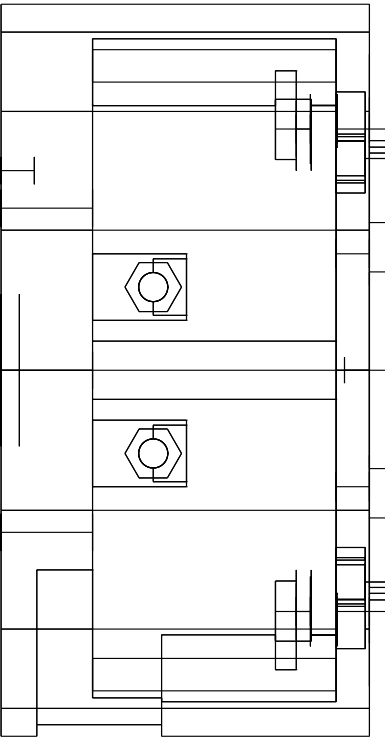
No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A13-IF-301-recessed-carrier

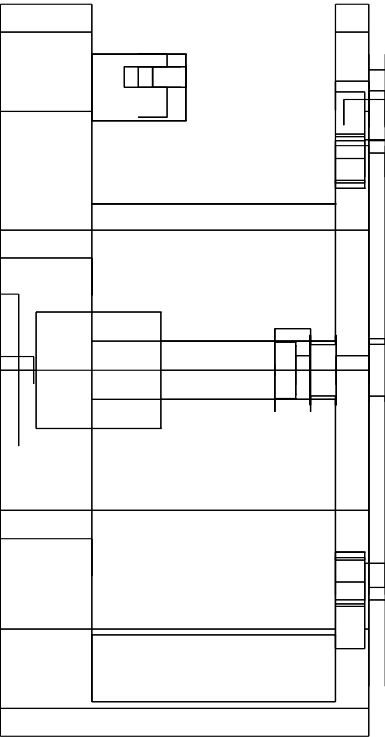
mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



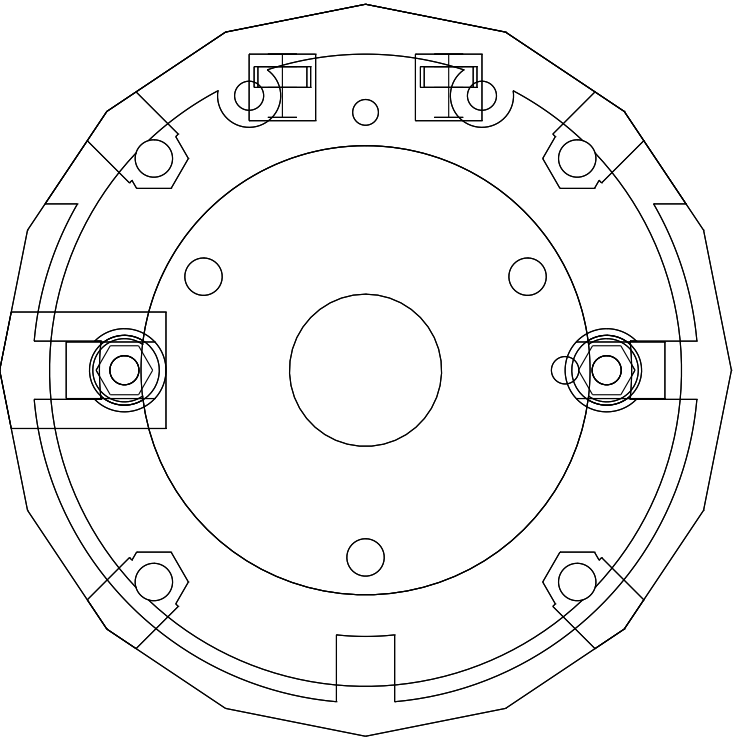
46.300 overall / drawing scale1.100
Vertical88.000; CAD axes XY

XZ all edges, hidden included



46.300 overall / drawing scale1.100
Vertical88.000; CAD axes XZ

YZ all edges, hidden included



88.000 overall / drawing scale1.100
Vertical88.000; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

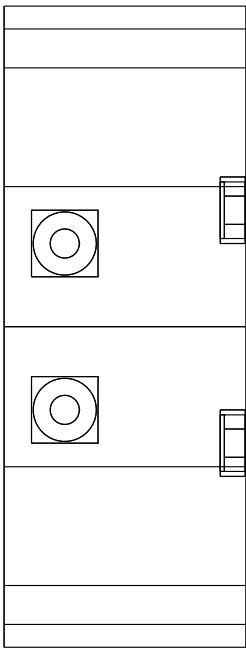
A13-IF-302-dorsal-cover

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

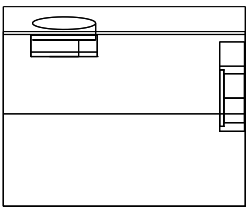
XY all edges, hidden included

XZ all edges, hidden included

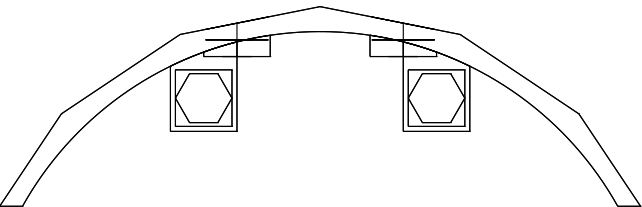
YZ all edges, hidden included



29.100 overall / drawing scale1.100
Vertical77.076; CAD axes XY



29.100 overall / drawing scale1.100
Vertical24.000; CAD axes XZ



77.076 overall / drawing scale1.100
Vertical24.000; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.

A13-IF-303-face-insert

mm | supported, unpowered, unloaded fit issue | STEP defines complete nominal geometry

XY all edges, hidden included



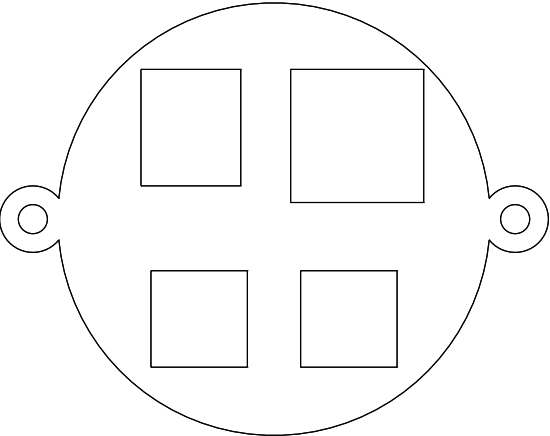
3.000 overall / drawing scale1.100
Vertical65.984; CAD axes XY

XZ all edges, hidden included



3.000 overall / drawing scale1.100
Vertical51.963; CAD axes XZ

YZ all edges, hidden included



65.984 overall / drawing scale1.100
Vertical51.963; CAD axes YZ

Retained original A13 cartridge/interface STEP, unmodified; source dimensions and bearing stack in linked fit documents.
J7 local mounting/shaft datums retained. Supplier motor pilot, bearing seats and captive nuts need physical gauges.

Printed default linear +/-0.2 after coupons; drilled holes +/-0.1 as trial targets; no metal GD&T release.

No inferred tapped plastic threads. Purchased motors/tubes/steel sleeves/fasteners are not printed.

Projection scale varies per view. Original CAD frame retained. Check supports, nut access, manual fit and actual motor plugs.